

BCA(H)¹ Seventh Semester Syllabus²

Introduction to Data Science: BCAC701A

MODULES	TOPICS	MARKS
1	Introduction To Statistics for Data Science: • Role Of Statistics in Data Science • Types Of Data and Variables • Population Vs Sample • Data Collection and Classification	10
2	Measures Of Central Tendency • Mean, Median, Mode • Geometric Mean, Harmonic Mean • Applications And Simple Problems	15
3	Measures Of Dispersion • Range, Quartile Deviation • Mean Deviation, Standard Deviation • Variance, Coefficient of Variation	15
4	Correlation And Regression • Karl Pearson Correlation • Spearman Rank Correlation • Regression Lines (X On Y, Y On X) • Interpretation	15
5	Probability And Distributions • Sample Space; Events; Types of Events • Axiomatic Approach to Probability • Problems Based on Permutation and Combination • Addition Theorem • Conditional Probability • Multiplication Theorem • Independent Events • Total Probability • Bayes' Theorem • Binomial Distribution and Its Use • Poisson Distribution and Its Use • Normal Distribution (<i>Basic Idea and Use</i>)	15
6	Basic Data Science Methods • Introduction To Classification • Decision Tree • Clustering • Simple Examples and Applications	30

¹ From 7th Semester it becomes **BCA HONOURS**

² Link to the Original Syllabus: https://makautexam.net/aicte_details/Syllabus4/BCA/sem723.pdf

Data Science Lab with Python: BCAC791A

MODULES	TOPICS
1	Python Fundamentals for Data Science: • Variables, Data Types • Lists, Tuples • Dictionaries • Loops • Functions
2	NumPy For Numerical Operations: • Array: Creation, Indexing, Slicing • Mathematical Operations on Arrays
3	Pandas For Data Handling: • Series And DataFrame • Importing CSV Files • Data Selection, Filtering
4	Data Cleaning and Preparation • Handling Missing Values • Removing Duplicates • Data Transformation
5	Exploratory Data Analysis and Visualization • Descriptive Statistics • Summary Measures • Using Matplotlib: Line, Bar Chart, Histogram, Pie Chart
6	Statistical Measures and Probability Using Python • Mean, Median, Mode • Standard Deviation • Variance • Simple Probability Examples
7	Basic Data Science Techniques: • Implementation Of: Classification, Decision Tree, Clustering on Simple Datasets • Mini Case Study

Suggested Reading:

1. **Data Science and Business Analytics** — S. P. Rajagopalan, Publisher: Wiley India
2. **Statistics for Data Scientists: 50 Essential Concepts** — Peter Bruce & Andrew Bruce, Publisher: O'Reilly (Indian edition available)
3. **Fundamentals of Statistics** — S. C. Gupta & V. K. Kapoor, Publisher: Sultan Chand & Sons
4. **Data Science: Introductory and Advanced Topics** — N. G. Das, Publisher: McGraw Hill Education India
5. **Machine Learning & Data Science – An Introduction to Statistical Learning Methods with R and Python** — Daniel D. Gutierrez, Publisher: CRC Press (Indian edition)
6. **Python for Data Science** — Dr. Basant Agarwal & Dr. Sumit Chauhan, Publisher: Oxford University Press India

Introduction to Machine Learning: BCAC701B

MODULE	TOPICS	MARKS
1	Introduction to ML & Data Preprocessing: • Meaning and scope of ML • Types of learning • Applications • ML workflow • Data types • Preprocessing • Handling missing values • Feature selection • Training and testing data	20
2	Regression Techniques: • Simple linear regression • Multiple regression (<i>basic idea</i>) • Model prediction • Evaluation using MAE, MSE, RMSE	20
3	Classification Techniques: • Logistic Regression • KNN • Naïve Bayes • Confusion matrix • Accuracy, precision, recall	20
4	Clustering and Decision Tree: • K-Means clustering • Centroid concept • Decision tree structure • Entropy and information gain • Tree construction	20
5	Model Evaluation and Applications: • Model validation • Overfitting and underfitting • Cross-validation (<i>basic idea</i>) • ML applications • Case study	20

Machine Learning Lab: BCAC791B

MODULE	TOPIC
1	Python Environment and Data Handling • Working In Jupyter/IDE • Using NumPy and Pandas • Importing CSV Datasets • Basic Data Operations
2	Data Preprocessing and EDA • Handling Missing Values • Data Cleaning • Feature Selection • Train-Test Split • Visualisation Using Matplotlib
3	Regression Techniques • Implementing Simple Linear Regression • Prediction • Calculating MAE, MSE, RMSE
4	Classification Techniques • Logistic Regression, KNN, Naïve Bayes • Confusion Matrix • Accuracy, Precision, Recall
5	Decision Tree and Clustering • Building A Decision Tree Model • Implementing K-Means Clustering • Visualisation And Interpretation
6	Model Evaluation and Mini Project • Cross-Validation (<i>Basic Idea</i>) • Overfitting And Underfitting • Applying Full ML Workflow on A Small Dataset

Suggested Reading:

1. **Machine Learning using Python** — Manaranjan Pradhan & U. Dinesh Kumar, Publisher: Wiley India (2019; also 2nd edition 2025)
2. **Machine Learning with Python** — Abhishek Vijayvargia, Publisher: BPB Publications (2018)
3. **Applied Machine Learning Solutions with Python** — Siddhanta Bhatta, Publisher: BPB Publications (2022)
4. **Introduction to Machine Learning with Python: A Guide for Data Scientists** — Andreas C. Müller & Sarah Guido (Greyscale Indian Edition), Publisher: Shroff Publishers (2016)
5. **Data Science and Machine Learning Using Python** — Reema Thareja, Publisher: McGraw Hill India

Research Methodology: BCAC702

MODULE	TOPIC	MARKS
1	Foundations of Research • Meaning, Objectives, And Types of Research: ○ Basic, ○ Applied, ○ Exploratory, Descriptive • Research Process and Characteristics of Good Research • Identifying And Defining a Research Problem • Formulation Of Research Questions and Objectives • Variables In Research and Research Motivation	15
2	Literature Review and Research Design: • Importance Of Literature Review • Sources Of Literature: Journals, Conference Papers, Reports, Online Databases • Research Gap Identification • Types Of Research Design: ○ Experimental, ○ Descriptive, ○ Analytical • Developing Hypotheses and Conceptual Framework	15
3	Sampling Techniques and Data Collection Methods: • Census Vs Sampling • Probability And Non-Probability Sampling Methods • Primary And Secondary Data Collection • Methods: Surveys, Observation, Interview, Questionnaire • Designing Questionnaires and Schedules	20
4	Statistical Tools for Research: • Classification And Tabulation of Data • Measures Of Central Tendency and Dispersion • Correlation And Regression Basics • Estimation Of Population Parameters • Hypothesis testing: t-test, chi-square test, ANOVA-concept	20
5	Data Analysis, Interpretation and Presentation • Data Coding and Data Processing • Use Of Charts, Tables, And Graphs for Presentation • Interpretation Of Statistical Results • Drawing Conclusions and Recommendations • Use Of Software Tools for Analysis (Overview)	15
6	Research Reporting and Ethics • Structure Of Research Report / Thesis / Paper • Writing Abstract, Introduction, Methodology, Results, Conclusion • Referencing styles: (APA/MLA/IEEE) • Plagiarism And Research Ethics • Preparing Research Proposal and Presentation	15

Research Methodology Lab: BCAC792

EXP NO.	EXPERIMENTS
1	Introduction to Research and Identification of Research Problem: Understand different types of research and formulate research problems
2	Literature Review using Digital Libraries: Search and review research papers using databases such as Google Scholar, IEEE Xplore, Springer, and ACM Digital Library
3	Research Objectives, Questions and Hypothesis Formulation: Develop research objectives, research questions, and hypotheses
4	Research Design and Sampling Techniques: Learn qualitative, quantitative, and mixed research methods along with sampling techniques
5	Designing Questionnaire using Google Forms: Create structured questionnaires for data collection
6	Data Collection and Data Cleaning Collect primary/secondary data and perform preprocessing
7	Statistical Analysis using MS Excel/SPSS/R Compute descriptive statistics such as mean, median, mode, standard deviation, and variance
8	Data Visualization Generate charts, graphs, and dashboards for data interpretation
9	Hypothesis Testing Perform t-test, chi-square test, ANOVA, or correlation analysis based on sample data
10	Citation and Reference Management Use tools such as Zotero or Mendeley for citations and bibliography in IEEE/APA formats
11	Plagiarism Checking and Ethical Research Practices Check similarity using plagiarism detection tools and understand research ethics
12	Research Paper Writing using IEEE/APA Format Prepare a complete research paper including Abstract, Introduction, Methodology, Results, Conclusion, and References

Reference Books:

1. **Research Methodology: Methods and Techniques** — C. R. Kothari, Publisher: New Age International Publishers
2. **Research Methodology: A Step-by-Step Guide for Beginners** — Ranjit Kumar, Publisher: Sage Publications India
3. **Research Methods for Business: A Skill-Building Approach** — Uma Sekaran & Roger Bougie, Publisher: Wiley India
4. **Fundamentals of Research Methodology and Statistics** — Yogesh Kumar Singh, Publisher: New Age International Publishers
5. **Business Research Methods** — Donald R. Cooper & Pamela S. Schindler, Publisher: McGraw Hill Education (India)
6. **Social Science Research: Principles, Methods & Practices** — Anol Bhattacharjee, Publisher: University of Florida Press (widely used in India)
7. **Research Design: Qualitative, Quantitative, and Mixed Methods Approaches** — John W. Creswell & J. David Creswell, Publisher: Sage Publications (International edition often available in India)

Cyber Security: BCAC703

MODULE	TOPICS	MARKS
1	Networking Fundamentals for Cyber Security • Introduction to Networking • OSI & TCP/IP Framework • Transmission Media (with Case Study) • Network Devices (with Case Study) • TCP/IP Framework in detail • Subnet Masking (with Case Study) • Three-way Handshaking • NAT • Different Protocols & their uses	20
2	Cyber Security Foundations • Cyber Security Definition • Security Triad (CIA) with Case Study • Hacking & Ethical Hacking • Types of Hackers • Phases of Hacking (<i>Reconnaissance, Scanning, Gaining Access, Maintaining Access, Covering Tracks</i>) • Cryptography & Watermarking and their use • Types of Cryptography • Symmetric vs Asymmetric Algorithms • RSA Algorithm explained	20
3	Information Security Management • Introduction to ISM • Goals of Information Security • ISMS (Information Security Management System) working procedure • Security Policies • Risk Management • Information Security Audit (with Case Study)	15
4	Cyber Attacks • Types of Attacks: Passive & Active Attacks • Malware • Phishing • Denial of Service • SQL Injection • Social Engineering • Insider Threats • Case Studies of Major Cyber Attacks	20
5	Technological Aspects of Cyber Security • Firewalls • IDS & IPS • VPNs • Security Protocols (<i>SSL, TLS, IPsec</i>)	25

	• Intrusion Detection Systems • Honeypots • Multi-Factor Authentication • Emerging Technologies in Cyber Security (<i>AI, Blockchain in Security, Quantum Cryptography</i>)	
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Books & References:

1. **Fundamentals of Cyber Security (Principles, Theory & Practices)** — Bhushan, M., & Rathore, R. S. (2017), Publisher: BPB Publications, New Delhi
2. **Cyber Security in India: Education, Research and Training** — Shukla, S. K., & Agrawal, M. (2020), Publisher: Springer, Singapore
3. **Cyber Security and Cyber Laws** — Basta, A., Basta, N., Brown, M., & Kumar, R. (2018), Publisher: Cengage Learning India, New Delhi
4. **Cyber Security** — Godbole, N. (2011), Publisher: Wiley India, New Delhi
5. **Cryptography and Network Security** — Mukhopadhyay, D., & Forouzan, B. A. (2011), Publisher: Tata McGraw-Hill Education, New Delhi

Consumer Behavior: MIM701A

MODULE	TOPICS	% of QA
1	Introduction To Consumer Behavior • Evolution of consumer behavior • Understanding consumers and market segments • Consumer behavior and marketing strategy • Psychographic dimensions • Consumer motivation • Perception • Personality • Information processing • Attitude formation and attitude change • Scope and their applications • Information search process • Evaluative criteria and decision rules • Building customer satisfaction • 7'Os of consumer behavior	10%
2	Factors Affecting Consumer Behavior • Factors influencing consumer behavior • External Influences ◦ Culture ◦ Sub Culture ◦ Social Class ◦ Reference Groups ◦ Family • Internal Influences ◦ Needs & Motivations ◦ Perception ◦ Personality ◦ Lifestyle ◦ Values ◦ Learning ◦ Memory ◦ Beliefs & Attitudes	10%
3	Social and Cultural Environment • Economic • Demographic • Cross Cultural and Socio-Cultural Influences • Cultural Relevance to Marketing Decisions • Characteristics of Culture • Cultural Values • Cultural Changes • Cross Cultural Understandings • Social Stratification • Reference Groups and Family Influences • Personal Influence • Family ◦ Role & Structure	20%

	○ Family Life Cycle ○ Purchasing Decisions ○ Changing Role of Families • Role of Reference Group • Types of Reference Group	
4	Consumers Need, Motivation, Personality, Perception and Learning • Motivation ○ Needs ○ Goals ○ Motive Arousal ○ Maslow Hierarchy of Needs ○ Alderfer's ERG Theory ○ Herzberg Motivation-Hygiene Theory ○ McClelland's Achievement Theory ○ McGregor's Theory X and Y ○ Vroom's Expectancy Theory ○ Porter's Expectancy Theory ○ Freud's Theory of Motivation • Personality ○ Meaning and Concept of Personality ○ Determinant of Personality ○ Theory of Personality ▪ Self-concept Theory ▪ Psychoanalytic Theory ▪ Neo-Freudian Theory ▪ Trait Theory ▪ Humanistic Theory ▪ Social-cognitive Theory ○ Measurement of Personality ○ Development of Personality • Perception ○ Definition and Importance ▪ Defining Perception ▪ Nature of Perception ▪ Significance in Consumer Behavior ○ The Perceptual Process ▪ Exposure ▪ Attention ▪ Organization ▪ Interpretation ▪ Retention ○ Elements of Perception ▪ Sensation ▪ Threshold ▪ Perceptual Biases ○ Sensory Dynamics ▪ Sight ▪ Sound ▪ Touch ▪ Smell	20%

	▪ Taste ○ Barriers to Accurate Perception ▪ Selective Attention ▪ Perceptual Defense ○ Perception and Consumer Decision-Making ▪ Perception of Products and Brands ▪ Perception of Price ▪ Perception of Risk ▪ Perception and Advertising ○ Perceptual Mapping ○ Subliminal Perception ○ Consumer Imagery • Learning ○ Meaning ○ Definition ○ Concept ○ Relationship with Consumer Behavior ○ Theories of Learning ▪ Behavioral Learning Theory ▪ Theory of Classical Conditioning (Stimulus-Response) ▪ Cognitive Associative Learning (Memory/Trial-Error Based) ▪ Operant or Instrumental Conditioning (Result/Reinforcement Based) ▪ Social Learning Theory ▪ Cognitive Theories ▪ Social Cognitive Theory ▪ Cognitive Behavioral Theory ▪ Constructivism	
5	Consumer Decision Making Process • Types of Consumer Decisions • Consumer Decision Making Process ○ Problem Recognition ○ Information Search ○ Alternative Evaluation ○ Purchase Selection ○ Post Purchase Evaluation • Buying Pattern in the New Digital Era • Four Views of Consumer Decision Rules ○ Economic Man ○ Passive Man ○ Emotional Man ○ Cognitive Man • Models of Consumer Decision Making ○ Nicosia Model	20%
6	Marketing Communications, Decision Making Models, Consumer Rights • Marketing Communication Process	20%

	• Types of Communication Systems ◦ Interpersonal ◦ Impersonal ◦ Persuasive Communication • Consumer Decision Making Models ◦ Black Box Model ◦ Economic Model ◦ Howard Model ◦ Howard-Sheth Model ◦ EKB Model ◦ Webster and Wind Model ◦ Sheth Industrial Buyer Behaviour Model • Consumer Protection Act 1986 • Rights of Consumers	
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Reading Suggestions:

- **Consumer Behaviour** – Anis Chattopadhyay – *Taurean Publications*
- **Consumer Behavior** – Kumar Leon G., Schiffman, Joe Wisenblit, S. Ramesh – *Pearson*
- **Consumer Behavior** – Schiffman – *Pearson*

MIM702?: Not Yet Selected

OPTIONS ARE:

1. STRATEGIC MANAGEMENT – MIM 702A
2. INTELLECTUAL PROPERTY RIGHTS – MIM 702B

LINK:https://makautexam.net/aicte_details/CourseStructure4/Minor_in_Management_chain_for_wef2324.pdf

NOT FINAL